

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Ratio Problem Solving

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

RATIO PROBLEM SOLVING · CH1 · L13



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The Map

This lesson collapses Ratio Problem Solving into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Find one share

TRIGGER *"work out", "find", a missing value*

02 Use total parts

TRIGGER *"probability", "random", "given that"*

03 Convert mixed information to parts

TRIGGER *"probability", "random", "given that"*

04 Check the order of the ratio

TRIGGER *"probability", "random", "given that"*

05 Split the ratio

TRIGGER *"probability", "random", "given that"*

06 The ratio is

TRIGGER *"probability", "random", "given that"*

BANK EVIDENCE 7 website questions, 7 checked solutions, 3 local PDFs read.

01 Find one share

TRIGGER *"work out", "find", a missing value*

BECOMES A Ratio Problem Solving question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\text{Brendon : Asha : Julie} = 3 : 2 : 6$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1HR Q1

The ratio is

$$\text{Brendon : Asha : Julie} = 3 : 2 : 6$$

Asha and Julie together have

$$2 + 6 = 8 \text{ parts}$$

Finish: Brendon receives \$13.50.

FORMULA BANK

Turn every sentence into shares or equations before calculating.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 01

SAME IDEA Use the same first line from Strategy 01.

QUESTION

1 Brendon, Asha and Julie share some money in the ratios 3 : 2 : 6 The total amount of money that Asha and Julie receive is \$36 Work out the amount of money that Brendon receives. \$ 1 5 2

YOUR SOLUTION

02 Use total parts

TRIGGER *"probability", "random", "given that"*

BECOMES A Ratio Problem Solving question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\text{Brendon : Asha : Julie} = 3 : 2 : 6$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1HR Q1

The ratio is

$$\text{Brendon : Asha : Julie} = 3 : 2 : 6$$

Asha and Julie together have

$$2 + 6 = 8 \text{ parts}$$

Finish: Brendon receives \$13.50.

FORMULA BANK

Turn every sentence into shares or equations before calculating.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 02.

QUESTION

1. Brendon, Asha and Julie share some money in the ratios $3 : 2 : 6$. The total amount of money that Asha and Julie receive is \$36. Work out the amount of money that Brendon receives. \$ 15.2

YOUR SOLUTION

03 Convert mixed information to parts

TRIGGER *"probability", "random", "given that"*

BECOMES A Ratio Problem Solving question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\text{Danil : Gabriel : Hadley} = 3 : 5 : 9$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2022 P1H Q2

The ratio is

$$\text{Danil : Gabriel : Hadley} = 3 : 5 : 9$$

Split the ratio

$$9 - 5 = 4 \text{ parts}$$

Finish: Danil receives 147 euros.

FORMULA BANK

Turn every sentence into shares or equations before calculating.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 03

SAME IDEA Use the same first line from Strategy 03.

QUESTION

2 Danil, Gabriel and Hadley share some money in the ratios 3 : 5 : 9 The difference between the amount of money that Gabriel receives and the amount of money that Hadley receives is 196 euros. Work out the amount of money that Danil receives. euros

YOUR SOLUTION

04 Check the order of the ratio

TRIGGER *"probability", "random", "given that"*

BECOMES A Ratio Problem Solving question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\text{Brendon : Asha : Julie} = 3 : 2 : 6$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1HR Q1

The ratio is

$$\text{Brendon : Asha : Julie} = 3 : 2 : 6$$

Asha and Julie together have

$$2 + 6 = 8 \text{ parts}$$

Finish: Brendon receives \$13.50.

FORMULA BANK

Turn every sentence into shares or equations before calculating.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

05 Split the ratio

TRIGGER *"probability", "random", "given that"*

BECOMES A Ratio Problem Solving question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\frac{7}{10} \times 20160 = 14112$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P1H Q1
of the money is changed

$$\frac{7}{10} \times 20160 = 14112$$

Split the ratio

$$\text{dollars} : \text{rupees} : \text{francs} = 9 : 5 : 2$$

Finish: 6174 rand more is changed into dollars than francs.

FORMULA BANK

Turn every sentence into shares or equations before calculating.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 05

SAME IDEA Use the same first line from Strategy 05.

QUESTION

1 Pieter owns a currency conversion shop. Last Monday, Pieter changed a total of 20 160 rand into a number of different currencies. He changed 3 10 of the 20 160 rand into euros. He changed the rest of the rands into dollars, rupees and francs in the ratios 9 : 5 : 2 Pieter changed more rands into dollars than he changed into francs. Work out how many more. rand

YOUR SOLUTION

06 The ratio is

TRIGGER *"probability", "random", "given that"*

BECOMES A Ratio Problem Solving question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\text{Brendon : Asha : Julie} = 3 : 2 : 6$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1HR Q1

The ratio is

$$\text{Brendon : Asha : Julie} = 3 : 2 : 6$$

Asha and Julie together have

$$2 + 6 = 8 \text{ parts}$$

Finish: Brendon receives \$13.50.

FORMULA BANK

Turn every sentence into shares or equations before calculating.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 06

SAME IDEA Use the same first line from Strategy 06.

QUESTION

1 Brendon, Asha and Julie share some money in the ratios 3 : 2 : 6 The total amount of money that Asha and Julie receive is \$36 Work out the amount of money that Brendon receives. \$ 1 5 2

YOUR SOLUTION

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "find", a missing value	Find one share
"probability", "random", "given that"	Use total parts
"probability", "random", "given that"	Convert mixed information to parts
"probability", "random", "given that"	Check the order of the ratio
"probability", "random", "given that"	Split the ratio
"probability", "random", "given that"	The ratio is

FINAL BOTTOM LINE Do not try to remember every past-paper wording. Remember the

trigger, write the first line, and let the working follow.